

PathFinder: Personalized Career and Learning Roadmaps for CS students

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Abstract—As the computer science field continues to evolve, new careers and skill requirements emerge rapidly, making it increasingly difficult for students to identify suitable career paths and remain aligned with industry trends. Students may pursue careers that appear promising but ultimately prove unsuitable due to a mismatch between their personality traits, interests, or technical skills. Furthermore, even when students identify an appropriate career path, they often struggle to find structured learning resources and a clear roadmap to achieve their goals. To address these challenges, we developed PathFinder, an intelligent career prediction and course recommendation platform that generates personalized learning roadmaps for computer science students. The system incorporates machine learning and Retrieval-Augmented Generation (RAG) technologies to predict the most suitable career path based on a student's personality and skills while providing targeted educational recommendations. The system employs two machine learning models, one trained on personality-related features and another on technical skills, which are combined using a multimodal late-fusion ensemble approach. This method enables both personality and skill signals to contribute to the final career prediction while identifying the student's skill gaps relative to the predicted career path. The prediction models are deployed through a FastAPI backend and integrated with a hybrid RAG-based recommendation engine. Using the predicted career path and identified skill gaps as input, the RAG engine retrieves relevant learning resources through semantic vector search and metadata-based filtering and leverages a large language model to generate personalized course recommendations and structured learning roadmaps. By combining career prediction, skill-gap identification, intelligent course retrieval, and AI-generated guidance within a single platform, PathFinder helps students make informed career decisions while providing a clear and personalized pathway for skill development.

Keywords—career prediction, ensemble learning, multimodal machine learning, skill-gap analysis, retrieval-augmented generation, course recommendation, personalized learning roadmap

I. INTRODUCTION

As the computer science field continues to evolve, new careers and skill requirements emerge rapidly, making it increasingly difficult for students to identify suitable career paths and remain aligned with industry trends. Research further shows that aligning career choices with a person's personality and interests is a measurable predictor of long-term job satisfaction and performance. Even when students do find a suitable career, they often struggle to keep up with the new technologies and knowledge it demands. These challenges highlight the need for intelligent systems that help students choose the right career by leveraging their personality and skills, and building a personalized learning and guidance experience tailored to each student. Although several career recommendation systems have been proposed, many rely exclusively on personality assessments or technical skills independently, resulting in incomplete career evaluations. Furthermore, most existing systems focus solely on career prediction and do not provide personalized learning pathways to help students address identified skill gaps. Consequently, students may still struggle to acquire the competencies required for their chosen careers despite receiving career recommendations. PathFinder is a web application that was developed to help address these limitations and provide CS students with a full career prediction and personalized learning guidance. The system allows the student to fill a survey form that consists of questions about the student's skills, performance, and personality traits. Then the system predicts the student's career and performs a skill gap analysis and provides a full roadmap for the user to

complete their missing skills. The system also automatically updates the roadmap as the student progresses.

II. BACKGROUND

Career recommendation systems have evolved from traditional rule-based approaches to intelligent machine learning solutions that analyze student characteristics and recommend suitable career paths. Early studies primarily relied on academic performance or technical skills as predictors, while more recent work has incorporated personality traits, behavioral attributes, and multimodal learning techniques to improve prediction accuracy.

Yaqoob et al. demonstrated that feature engineering significantly improves career prediction by selecting the most informative features before classification. Their hybrid framework achieved approximately 95% prediction accuracy but relied solely on structured student attributes without considering personality dimensions.

Faruque et al. proposed a career prediction system specifically targeting Computer Science students using NLP-based skill representations and several machine learning classifiers. Support Vector Machines achieved the best performance with an accuracy of 88.63%. Although the work established a strong benchmark, it relied exclusively on technical skills and ignored personality characteristics.

El-Keiey et al. introduced a feature-selection framework for Information Technology career prediction using academic, behavioral, and personality features. Their Random Forest classifier achieved approximately 90% accuracy, confirming that carefully selected features substantially improve prediction performance. However, all features were processed within a single model without separating heterogeneous information sources.

Research has also shown the effectiveness of multimodal ensemble learning. Jain et al. demonstrated that combining multiple independent prediction models through weighted fusion improves overall prediction accuracy compared to individual models. Similarly, Large et al. showed that heterogeneous ensembles consistently outperform single classifiers by combining calibrated probability estimates rather than raw predictions.

In parallel, course recommendation systems have shifted from keyword matching toward Retrieval-Augmented Generation (RAG). Traditional approaches used cosine similarity over bag-of-words representations, limiting semantic understanding. Recent systems such as RAMO utilize RAG to retrieve relevant educational resources while grounding large language model responses in external knowledge bases. However, most existing systems either lack quantitative evaluation, ignore explicit skill-gap information, or operate independently from career prediction modules.

PathFinder addresses these limitations by integrating two independently trained machine learning models through a multimodal late-fusion ensemble and combining them with a hybrid RAG-based recommendation engine. Unlike previous approaches, the system jointly performs career prediction, skill-gap identification, semantic course

retrieval, personalized roadmap generation, and continuous progress tracking within a unified platform.

TABLE I. Comparison of Existing Career Prediction Systems

Study	Input Features	Prediction Method	Limitation
Yaqoob et al.	Selected academic features	Hybrid MLP	No personality analysis
Faruque et al.	Technical skills	SVM	Skills only
El-Keiey et al.	Academic + personality	Random Forest	Single-model approach
Jain et al.	Multimodal signals	Ensemble Learning	Employment matching only
PathFinder	Personality + Skills	Late-Fusion Ensemble + RAG	Integrated end-to-end system

III. PROPOSED SYSTEM

PathFinder is an intelligent career guidance platform that integrates machine learning and Retrieval-Augmented Generation (RAG) to provide personalized career recommendations and learning roadmaps for Computer Science students. Unlike conventional systems that focus solely on career prediction, PathFinder combines personality analysis, technical skill assessment, skill-gap identification, and personalized learning recommendations within a unified framework.

The system follows a layered architecture consisting of a web-based client, a FastAPI backend, and a MongoDB database. The backend contains two cooperating intelligent modules: the Career Prediction Module and the Course Recommendation Module. The Career Prediction Module estimates the student's most suitable career by combining two independently trained machine learning models, while the Course Recommendation Module retrieves relevant educational resources and generates a structured roadmap based on the predicted career and identified skill gaps.

The overall workflow of PathFinder is illustrated in Fig. 1. A student first completes an assessment containing questions related to personality traits, technical skills, academic performance, and learning preferences. The collected information is processed by the prediction module to determine the most suitable career path. The predicted career and detected skill gaps are then supplied to the

RAG-based recommendation module, which retrieves relevant learning resources using semantic search and metadata filtering before generating a personalized learning roadmap.

The system further supports progress tracking by continuously monitoring completed courses and acquired skills. As the student's profile evolves, recommendations and learning roadmaps are regenerated to ensure alignment with the student's current capabilities and career objectives.

IV. METHODOLOGY

A. Career Prediction Module

The Career Prediction Module identifies the most suitable career path by jointly analyzing the student's personality traits and technical skills. Instead of relying on a single classifier, the proposed approach employs two independently trained machine learning models that capture complementary information about the student.

The first model focuses on personality characteristics together with selected academic attributes, while the second analyzes the student's technical competencies. After preprocessing and feature extraction, both models generate probability distributions over six predefined Computer Science career categories. These probabilities are calibrated and combined using a weighted late-fusion ensemble to produce the final career prediction.

Besides predicting the most appropriate career, the module constructs a skill-gap profile by comparing the student's current skills against the competency requirements of the predicted career. The generated skill-gap profile serves as the primary input to the recommendation module.

B. Course Recommendation Module

The Course Recommendation Module converts the predicted career and identified skill gaps into personalized educational recommendations through a hybrid Retrieval-Augmented Generation (RAG) pipeline.

A structured query is first generated using the predicted career, missing skills, learning preferences, and study constraints. Candidate courses are then retrieved from a curated knowledge base using semantic similarity search over dense vector embeddings. Metadata filtering further refines the candidate set according to course difficulty, duration, provider, and prerequisite requirements.

The retrieved courses are subsequently re-ranked using a hybrid scoring strategy that combines semantic similarity, skill-gap coverage, course quality, and metadata relevance. Finally, a Large Language Model generates personalized explanations describing why each recommended course is appropriate for the student.

C. Adaptive Learning Roadmap Module

Rather than presenting recommended courses as an unordered list, PathFinder automatically organizes them into a structured learning roadmap.

Using the ranked recommendations, identified skill gaps, and the student's available study time, the roadmap module divides learning into progressive phases beginning with foundational concepts and gradually advancing toward specialized career skills. Dependencies between technologies and prerequisite knowledge are considered when constructing the learning sequence, ensuring that students develop competencies in a logical order.

Each roadmap phase contains learning objectives, recommended courses, expected outcomes, and estimated completion durations, providing students with a clear pathway toward their target career.

D. Progress Tracking Module

The Progress Tracking Module continuously monitors the student's learning progress after roadmap generation.

As students complete recommended courses and update their technical skills, the system recalculates skill proficiency levels and career readiness. The learning roadmap is dynamically updated to reflect newly acquired competencies, while additional recommendations are generated whenever new skill gaps emerge.

The module also computes readiness indicators based on completed learning activities, mastered skills, and portfolio development, allowing students to monitor their overall progress toward becoming industry-ready professionals.

V. IMPLEMENTATION

PathFinder was implemented as a full-stack web application integrating machine learning services with a Retrieval-Augmented Generation backend.

The frontend was developed using **React** and **TypeScript**, providing an interactive interface for career assessment, recommendation visualization, roadmap management, and progress tracking.

The backend was implemented using **FastAPI**, exposing RESTful APIs responsible for authentication, machine learning inference, recommendation generation, and roadmap management.

Machine learning models were developed using **Scikit-learn**, **LightGBM**, **XGBoost**, and **Optuna**, while **Sentence Transformers** and **ChromaDB** were employed for semantic retrieval. **Google Gemini** was integrated to generate personalized learning roadmaps from the retrieved course set.

All application data—including user profiles, assessments, recommendations, roadmaps, and progress records—are stored in **MongoDB**, enabling efficient document-oriented storage and scalable system deployment.

The modular implementation allows each intelligent component to be updated independently, facilitating future improvements to prediction models and recommendation

algorithms without affecting the overall system architecture.

VI. RESULTS AND DISCUSSION

The proposed PathFinder system was evaluated by assessing the performance of both career prediction models and validating the effectiveness of the integrated recommendation pipeline. Experiments were conducted using separate datasets for the personality and skills models, with hyperparameter optimization performed using Optuna to maximize predictive performance.

A. Career Prediction Performance

Multiple machine learning algorithms were evaluated for both prediction tasks before selecting the final models. For the personality-based prediction, LightGBM outperformed Decision Tree, Support Vector Machine, Random Forest, K-Nearest Neighbors, and Naïve Bayes after hyperparameter optimization. The final model achieved an accuracy of **81%** with a weighted F1-score of **0.81**.

Similarly, the skills-based prediction model compared Logistic Regression, Random Forest, and XGBoost. XGBoost consistently achieved the highest performance, obtaining an accuracy and weighted F1-score of **91.7%** after Optuna optimization.

Table II summarizes the final performance of both selected models.

Table II Performance of the Career Prediction Models

Component	Selected Algorithm	Accuracy	F1-S core
Personality Analysis	LightGBM	0.81	0.81
Skill Analysis	XGBoost	0.917	0.917

The superior performance of the skills model indicates that technical competencies provide stronger predictive signals for career classification. Nevertheless, personality traits contribute complementary information that cannot be captured by technical skills alone. Consequently, combining both models through the proposed late-fusion ensemble enables more comprehensive career recommendations than either model individually.

B. Skill-Gap Identification

Following career prediction, the system identifies the gap between the student's current competencies and the expected skill profile of the predicted career. Career profiles were constructed during training by analyzing the most representative skills within each career category. The generated reports classify skills into mastered, underdeveloped, and missing competencies.

This information provides a structured representation of the student's strengths and weaknesses, allowing subsequent recommendation stages to prioritize learning resources that directly address the identified deficiencies.

C. Course Recommendation Evaluation

The recommendation module was evaluated qualitatively by examining the relevance of retrieved courses to the predicted career and identified skill gaps. The hybrid retrieval strategy successfully combined semantic similarity with metadata filtering to retrieve educational resources closely aligned with the student's learning objectives.

Unlike conventional keyword-based recommendation systems, semantic vector retrieval enabled the discovery of conceptually related courses even when exact skill names were absent. Metadata filtering further improved recommendation quality by considering factors such as course difficulty, duration, prerequisites, and educational provider.

The integration of Retrieval-Augmented Generation (RAG) allowed the large language model to generate personalized explanations and structured learning plans while grounding its responses in retrieved course documents, thereby reducing hallucinations and improving recommendation reliability.

D. Discussion

The experimental results demonstrate that separating personality analysis and technical skill assessment into independent machine learning models provides several advantages over a single unified classifier. Each model specializes in learning from a homogeneous feature space, resulting in improved predictive performance and greater interpretability.

The weighted late-fusion ensemble further enhances prediction quality by leveraging the complementary strengths of both models. Assigning greater weight to the skills model reflects its higher predictive accuracy while preserving the influence of personality traits, which are strongly associated with long-term career satisfaction and work preferences.

Furthermore, integrating career prediction with skill-gap analysis and Retrieval-Augmented Generation extends the functionality beyond traditional recommendation systems. Rather than simply predicting an appropriate career, PathFinder provides students with actionable learning guidance through personalized roadmaps that adapt as their skills evolve. This integrated approach supports continuous professional development and enables students to progress systematically toward their target careers.

VII. LIMITATIONS AND FUTURE WORK

Although PathFinder demonstrates promising results, several limitations remain.

First, the personality model was trained using publicly available datasets that may not fully represent the diversity of Computer Science students across different educational institutions and cultural backgrounds. Collecting larger datasets from real users would likely improve model generalization and prediction accuracy.

Second, the current recommendation knowledge base primarily contains online learning resources. Incorporating additional educational content such as university courses, technical documentation, open-source projects, coding challenges, certifications, and internship opportunities would provide more comprehensive learning guidance.

Third, the weighting strategy employed in the late-fusion ensemble is currently predefined based on experimental observations. Future work could investigate adaptive or learned weighting mechanisms that dynamically adjust each model's contribution according to prediction confidence and user characteristics.

Future development will also focus on integrating labor market analytics to align recommendations with current industry demand. Additional features such as portfolio evaluation, resume analysis, interview preparation, and automatic skill extraction from GitHub repositories could further enhance career guidance. Finally, longitudinal evaluation involving real students will be conducted to assess the long-term impact of personalized roadmaps on learning outcomes and career readiness.

VIII. CONCLUSION

This paper presented **PathFinder**, an intelligent career guidance platform that integrates machine learning and Retrieval-Augmented Generation to provide personalized career recommendations and adaptive learning roadmaps for Computer Science students.

Unlike conventional career recommendation systems that rely exclusively on personality assessments or technical skills, PathFinder combines independently trained personality and skills models through a multimodal late-fusion ensemble to produce more comprehensive career predictions. The system further performs automated skill-gap identification and employs a hybrid RAG-based recommendation engine to retrieve relevant educational resources and generate structured learning roadmaps tailored to each student's needs.

Experimental results demonstrated that the selected LightGBM and XGBoost models achieved competitive prediction performance, while the integrated recommendation pipeline successfully transformed career predictions into actionable learning plans. By combining career prediction, personalized course recommendation, roadmap generation, and continuous progress tracking within a single platform, PathFinder offers a practical solution for supporting Computer Science students in making informed career decisions and systematically developing the competencies required for their chosen professions.

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